



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: DATA SCIENCE (24AD3107)

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TITLE OF THE PROJECT	FinSight: Explainable Credit Risk Analytics and Loan Decision Support System	
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Abstract

Credit risk assessment helps financial institutions evaluate loan applicants and identify the possibility of repayment failure. Incorrect credit-risk classifications may lead to financial loss or the rejection of creditworthy applicants. In addition to predictive performance, credit-risk systems should provide understandable explanations so that model outputs can be reviewed effectively. This project proposes **FinSight**, a cloud-enabled explainable credit-risk analytics and decision-support system that integrates data preprocessing, machine learning, explainable artificial intelligence, cloud analytics, and an interactive web application.

The project uses the **UCI Statlog German Credit dataset**, which contains **1,000 applicant records with 20 input attributes and a good/bad credit-risk target**. The attributes include credit history, loan duration, credit amount, savings, employment, age, housing, and other applicant-related information. The original dataset is stored locally under the project's raw-data folder and is also uploaded to **Amazon S3** for cloud storage. Data validation checks the dataset structure, missing values, duplicate records, target labels, and categorical values. Exploratory data analysis is performed to study feature distributions, class distribution, and relationships between applicant characteristics and credit risk.

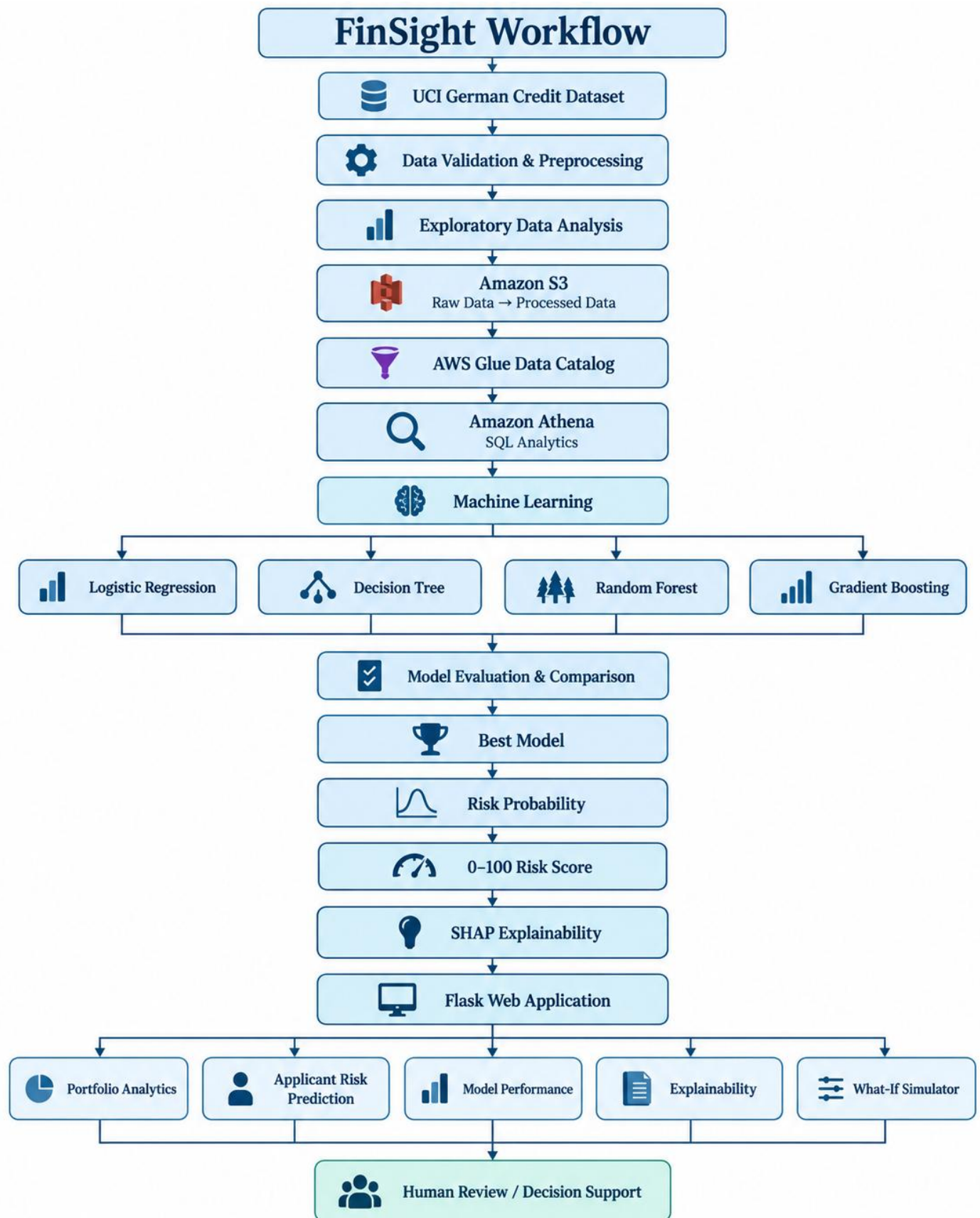
A reusable preprocessing pipeline is developed using Python and Scikit-learn. Categorical attributes are encoded and numerical features are transformed where required. The dataset is divided into stratified training and testing sets while ensuring that preprocessing is fitted only on training data to prevent data leakage. Four machine-learning algorithms — **Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting** — are trained and compared using **Accuracy, Precision, Recall, F1-score, ROC-AUC, confusion matrices, and cross-validation**. Evaluation also considers the unequal cost of classification errors, particularly the higher consequence of predicting a high-risk applicant as low risk.

The selected model produces the **probability of bad credit risk**, which is converted into a simple **0–100 educational risk score** and categorized into **Low, Moderate, or High predicted-risk levels**. **SHAP Explainable AI** is used to identify the important factors influencing overall model behaviour as well as individual applicant predictions. A **What-If Risk Simulator** allows selected applicant attributes to be modified so that changes in the model's predicted probability and risk score can be observed as model-sensitivity analysis.

For cloud-based analytics, **Amazon S3** is used to store raw and processed datasets, **AWS Glue** is used to catalog the processed data, and **Amazon Athena** is used to perform SQL-based analysis directly on the data stored in S3. A **Flask-based web application**, developed using HTML, CSS, Bootstrap, and Plotly, provides portfolio analytics, applicant risk prediction, risk scoring, explainability, model-performance comparison, and what-if analysis.

The expected outcome of FinSight is a reproducible end-to-end data-science application that demonstrates the complete workflow from raw-data preparation and cloud storage to machine-learning prediction and model interpretation. The system is intended as an **educational decision-support prototype** and does not automatically approve or reject loans. Instead, it provides transparent analytical outputs that can support human review and understanding of credit-risk predictions.

Project Architecture/Project Overview:



Signature of the Guide

HOD